

Richard Bonichon

PROTOCOL & FORMAL METHODS ENGINEER · PHD

✉ richard@loopless.xyz
☎ +33 7 81 84 10 24
📍 Saint-Alban, Occitanie, France

🌐 linkedin/richard-bonichon
🐙 github.com/rbonichon
🌐 rbonichon.github.io

- BFT CONSENSUS
- BLOCKCHAIN PROTOCOLS
- ZK PROOF SYSTEMS
- FUNCTIONAL PROGRAMMING (OCAML, RUST)
- FORMAL VERIFICATION & SMT
- PROGRAM ANALYSIS
- LLM-ASSISTED DEVELOPMENT

Protocol engineer with a formal-methods background: led the consensus team that shipped Tenderbake (Tezos BFT consensus), ZK and protocol work at O(1) Labs / Mina, now a Rust L1. Before that, 11 years at CEA leading the BINSEC binary-analysis platform; PhD in automated theorem proving.

WORK EXPERIENCE

2025-03	now	Senior Software Engineer · Repyh Labs Protocol Engineer - Documented the SDK and base libraries for early app developers - Modularized domain tasks; fixed transaction-application logic; typed abstractions - Added NFT support; designed and built the archive node - Built the integration-testing API and its test suite	Rust Rust Rust Rust
2022-09	2024-08	Software Engineer · O(1) Labs Crypto & Protocol Engineer - Codebase-wide cleanup, refactoring and higher test coverage - o1vm/Optimism: low-level comms with the fault-proof op-program - Berkeley hard-fork: pre/post-fork test harness - Prototyped account deletion (core zkApps feature)	OCaml Nix Rust Go Bash OCaml Dhall OCaml Postgresql
2021-01	2022-08	Software Engineer · Nomadic Labs Consensus Team Lead & Protocol Engineer - Test maintenance and migration Python → Text for the new consensus - Implemented Tenderbake consensus for the Ithaca protocol upgrade - Led the consensus team (from July 2021); Merge Team (core reviewers) member	Python OCaml OCaml
2019-09	2021-01	Senior Software Engineer · Tweag Project lead & Consultant - Tezos clients: Sapling, mockup, proxy - Gerrit plugins: owner approval, CI reporting - Built the internship program (setup, outreach, selection)	OCaml Rust Java JavaScript Go
2008-10	2019-08	Research Engineer · CEA Formal program analyses on source and binary code for safer, more secure software. - Lead dev & team co-lead of the open-source binary-analysis and symbolic-execution platform BINSEC - Co-advisor (day-to-day supervisor) of 2 PhD students (ICSE'21 Distinguished Paper, RAID'20) - Lectures on software attacks & vulnerabilities (ENSTA)	OCaml Docker Python
2013-07	2015-06	Visiting Professor · DIMAp/UFRN Teaching (programming, formal methods) / Research (SMT solvers) - SMTpp : generic pre-processor & analyzer for SMT solvers - SMT solver veriT development	OCaml C OCaml
2007	2008	Post-doc · LORIA / Paréo Generation of construction functions respecting algebraic invariants for OCaml	
2006	2007	ATER (Teaching assistant) · Université Paris 6 Pierre & Marie Curie	

PUBLICATIONS

22 peer-reviewed papers, including **ICSE 2021** (Distinguished Paper Award), **CCS 2021**, **ASE 2019**, **CAV 2018**, **RAID 2020**, **ACSAC 2019**. Full list: rbonichon.github.io/biblio.html

EDUCATION

2003	2006	PhD in Computer Science · Université Paris 6 Pierre & Marie Curie Thesis: <i>Tableaux et déduction modulo</i> Topics: automated theorem proving, rewriting, cut elimination, logic
2002	2003	MS in Computer Science (DEA) · Université Paris 7 Denis Diderot Programming: Semantics, Proofs & Logic
1999	2002	MS in Computer Science & Applied Mathematics · ENSEEIHT